13 SOUND

Sound waves are longitudinal waves and not transverse waves like light. Nevertheless they can be reflected and refracted in just the same way. In this chapter you will learn about the nature and behaviour of sound waves, and how we make use of them in our everyday lives ... and not just when we talk!



▲ Figure 13.1 Using sound waves to communicate.

LEARNING OBJECTIVES

- Know that sound waves are longitudinal waves which can be reflected and refracted

PHYSICS ONLY

- Know that the frequency range for human hearing is 20–20 000 Hz
- Practical: investigate the speed of sound in air
- Understand how an oscilloscope and microphone can be used to display a sound wave
- Practical: investigate the frequency of a sound wave using an oscilloscope
- Understand how the pitch of a sound relates to the frequency of vibration of the source
- Understand how the loudness of a sound relates to the amplitude of vibration of the source

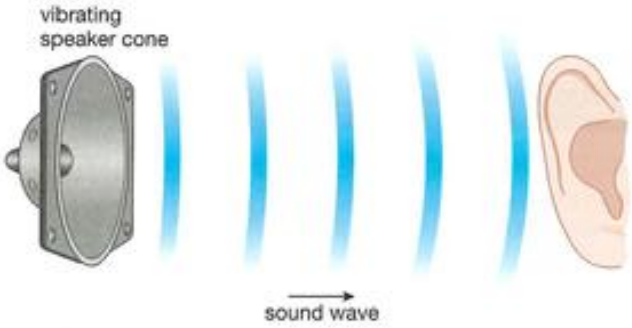


▲ Figure 13.2 The sound produced by the speakers must be loud but also of good quality.

Figure 13.2 shows part of the sound system used by a band playing at a concert. This equipment must produce sounds that are loud enough to be heard by all the audience and the sound quality must be good enough for the music to be appreciated. In this chapter we are going to look at how sounds are made and how they travel as waves.

SOUND WAVES

Sounds are produced by objects that are vibrating. We hear sounds when these vibrations, travelling as sound waves, reach our ears.



▲ Figure 13.3 The loudspeaker vibrates and produces sound waves.

EXTENSION WORK

You will not be tested on this more detailed description of how sounds from a loudspeaker are heard. As the speaker cone moves to the right, it pushes air molecules closer together, creating a **compression**. These particles then push against neighbouring particles so that the compression appears to be moving to the right. Behind the compression as the speaker cone moves to the left is a region where the particles are spread out. This region is called a rarefaction. After the cone has vibrated several times, it has created a series of compressions and rarefactions travelling away from it. This is a longitudinal sound wave (see page 98). When the waves enter the ear, they strike the eardrum and make it vibrate. These vibrations are changed into electrical signals, which are then detected by the brain.

PHYSICS ONLY

MEASURING THE SPEED OF SOUND

ACTIVITY 1

▼ PRACTICAL: INVESTIGATE THE SPEED OF SOUND USING ECHOES

A hinged 'clapper board' works best but it needs exterior handles to prevent fingers being trapped.

You can measure the speed of sound using two bits of wood and a stopwatch. Stand some distance away from a friend, start the stopwatch when you see her bang the two bits of wood together and stop the stopwatch when you hear the sound. Now using the equation speed, $v = \frac{\text{distance, } s}{\text{time, } t}$ you can calculate the speed of sound. The answer you are likely to get will not be very accurate unless you have very good eyesight and very fast reactions.

You can obtain a more accurate measurement using echoes (sounds that are reflected off a surface).

1. Both you and your friend should stand at least 50 m away from a large wall or building.
2. Bang the two pieces of wood together and listen for the echo.
3. Bang the pieces of wood together each time you hear the echo. This will create a regular rhythm of claps.
4. Ask the friend to time you doing 20 claps. During this time the sound will have travelled, for example, 50 m × 20 × 2 (to the wall and back 20 times), and you can divide this distance by the time to work out the speed of the sound.

The speed of sound in air is approximately 340 m/s, although this value does vary a little with temperature.

EXAMPLE 1

A girl stands 100 m from a tall building clapping her hands each time she hears an echo of her clap. It takes 11.7 s for her to hear 20 echoes. Calculate the speed of sound.

$$v = \frac{s}{t}$$

$$= \frac{100\,\text{m} \times 2 \times 20}{11.7\,\text{s}}$$

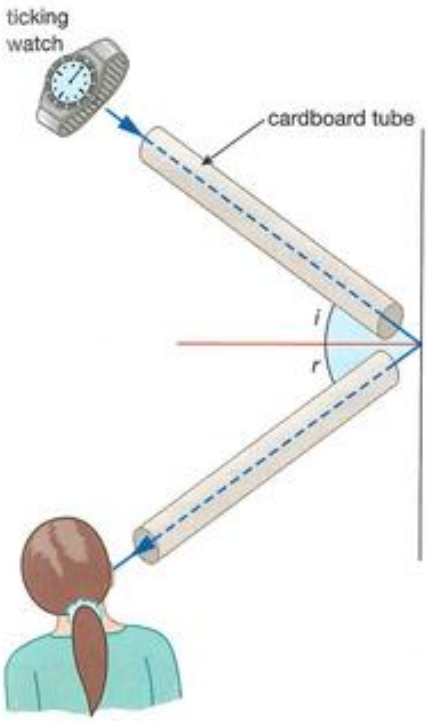
$$= 342\ \text{m/s}$$

END OF PHYSICS ONLY

REFLECTION

Sound waves behave in the same way as any other wave.

When a sound wave strikes a surface it may be reflected. Like light waves, sound waves are reflected from a surface so that the angle of incidence is equal to the angle of reflection (see Figure 13.4).



▲ Figure 13.4 Sound waves are reflected in the same way that light rays are reflected.

Ships often use echoes to discover the depth of the water beneath them. This is called echo sounding.

1. Sound waves are emitted from the ship and travel to the seabed (sea floor).
2. Some of these waves are reflected from the seabed back up to the ship.
3. Equipment on the ship detects these sound waves.
4. The time it takes the waves to make this journey is measured.
5. Knowing this time, the depth of the sea below the ship can be calculated.

The system of using echoes in this way is called **sonar** (Sound, Navigation And Ranging).

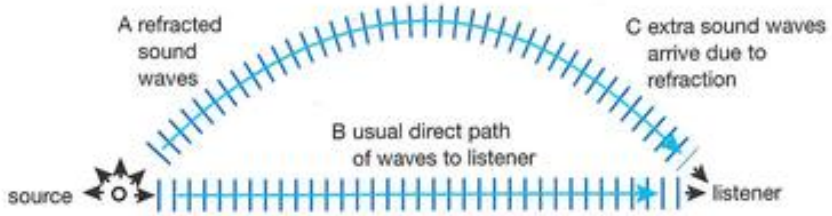


▲ Figure 13.5 Reflected sound can be used to tell ships about the depth of the sea beneath them.

REFRACTION OF SOUND

All waves can be refracted, even sound waves! For example, if some parts of a sound wave are travelling through warm air, they will travel more quickly than those parts travelling through cooler air. As a result the direction of the sound wave will change. It will be refracted.

Although it is not possible to see sound waves being refracted, we can sometimes hear their effect. Standing at the edge of a large pond or lake we can sometimes hear sounds from things on the other side of the water much more clearly than we would expect. This is due to refraction. Figure 13.6 explains how this happens.



▲ Figure 13.6 Why sometimes sounds travelling across water are louder than we expect

1. Most of the sound we hear travels to us in a straight line (Path B).
2. But some sound travels upwards (Path A).
3. If the temperature conditions are right, then as the sound waves travel through the air they are refracted and follow a curved path downwards (Path C).
4. We now receive two sets of sound waves.
5. So the sound we hear seems louder and clearer.

PHYSICS ONLY

PITCH AND FREQUENCY

Small objects, such as the strings of the violin in Figure 13.7, vibrate quickly and produce sound waves with a high frequency. These sounds are heard as notes with a high pitch.

Larger objects, such as the strings of the cello, vibrate more slowly and produce waves with a lower frequency. These sounds have a lower pitch.

The frequency of a source is the number of complete vibrations it makes each second. We measure frequency in hertz (Hz). If a source has a frequency of 50 Hz this means that it vibrates 50 times each second and therefore produces 50 waves each second.

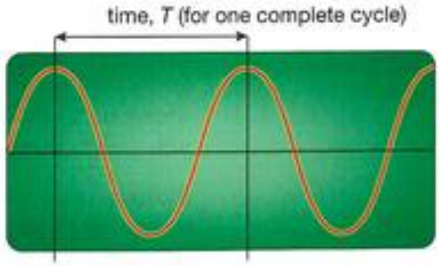


▲ Figure 13.7 The violin produces notes that are higher pitched than those from the cello.

ACTIVITY 2

▼ PRACTICAL: INVESTIGATE THE FREQUENCY OF A SOUND WAVE USING AN OSCILLOSCOPE

Although we cannot see an actual sound wave, we can see an image or representation of it by connecting a microphone to a piece of apparatus called an **oscilloscope**. When the sound wave from a source such as a tuning fork or vibrating string enters the microphone, the oscilloscope 'draws' the longitudinal sound wave as a transverse wave which allows us to see features such as the wave's amplitude and frequency more easily.



▶ Figure 13.8 An oscilloscope image of a sound wave



low-pitched sound high-pitched sound

▲ Figure 13.9 Low-pitched sounds and high-pitched sounds seen on an oscilloscope. A low-pitched sound has a low frequency, meaning fewer complete waves per second, so fewer complete waves are seen on the oscilloscope than for the high-pitched (high frequency) sound.

From the trace drawn on the screen we can measure the time for one complete vibration or one complete wave. This is called the time period of the wave (T). We can then find the frequency of the sound (f) using the equation:

$$f = \frac{1}{T}$$

EXAMPLE 2

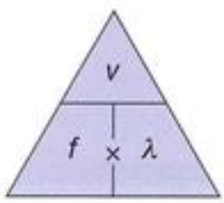
The picture of a sound wave seen on an oscilloscope has a time period of 0.005 s. What is the frequency of the wave?

$$f = \frac{1}{T}$$
$$= \frac{1}{0.005\ s}$$
$$= 200\ \text{Hz}$$

The relationship between the frequency (f), wavelength (λ) and speed of a wave (v) is described by the equation:

speed, v (m/s) = frequency, f (Hz) × wavelength, λ (m)

$$v = f \times \lambda$$



▲ Figure 13.10 The equation triangle for speed, frequency and wavelength.

EXAMPLE 3

Calculate the wavelength of a sound wave that is produced by a source vibrating with a frequency of 85 Hz. The speed of sound in air is 340 m/s.

$$v = f \times \lambda$$
$$340\ \text{m/s} = 85\ \text{Hz} \times \lambda$$
$$\lambda = \frac{340\ \text{m/s}}{85\ \text{Hz}}$$
$$= 4\ \text{m}$$

The wavelength is 4 m.

AUDIBLE RANGE

The average person can only hear sounds that have a frequency higher than 20 Hz but lower than 20 000 Hz. This spread of frequencies is called the audible range or hearing range. The size of the audible range varies slightly from person to person and usually becomes narrower as we get older. You can demonstrate this range by using a signal generator and a loudspeaker to produce sounds at different frequencies.

EXTENSION WORK

You will not be tested on ultrasounds and infrasounds but it is interesting to know what they are.
Some objects vibrate at frequencies greater than 20 000 Hz. The sounds they produce cannot be heard by human beings and are called ultrasounds. Some objects vibrate so slowly that the sounds they produce cannot be heard by human beings. These are called infrasounds.





▲ Figure 13.11 Some animals can make and hear sounds that lie outside the human audible range. Dolphins can communicate using ultrasounds. Elephants can communicate using sounds that have frequencies too low for us to hear (infrasounds).



▲ Figure 13.12 If you hit a drum hard, you get a louder sound than if you beat it gently.

LOUDNESS

If the drum in Figure 13.12 is hit hard, lots of energy is transferred to it from the drum stick. The drum skin vibrates up and down with a large amplitude, creating sound waves with a large amplitude and we hear a loud sound. If the drum is hit more gently less energy is transferred and sound waves with a smaller amplitude are produced. We hear these as quieter sounds.



▲ Figure 13.13 Loud sounds and quiet sounds on an oscilloscope



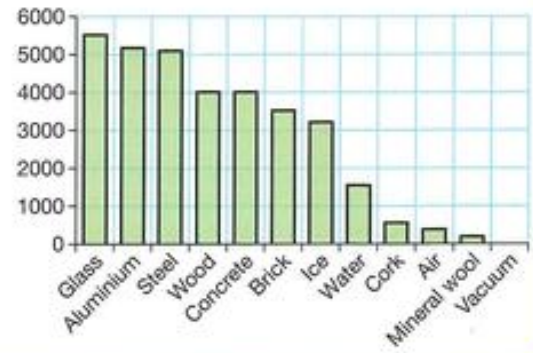
▲ Figure 13.14 A scientist measuring the sound level as a car passes him

LOOKING AHEAD – SOUND WAVES IN DIFFERENT MATERIALS

Sound waves can travel through:

- solids – this is why you can hear someone talking in the next room, even when the door is closed
- liquids – this is why whales can communicate with each other when they are under water
- gases – the sound waves we create when we speak travel through gases (in the air).

Sound waves cannot travel through a vacuum because there are no particles to carry the vibrations.



▶ Figure 13.15 Graph of speeds of sounds (in m/s) for different materials

CHAPTER QUESTIONS

More questions on sound and vibrations can be found at the end of Unit 3 on page 130.

SKILLS CRITICAL THINKING (4)

1 a Name a musical instrument that is used to produce high-pitched notes.

SKILLS REASONING (5)

b Explain why the musical instrument you have named in part a produces high-pitched notes.

(4)

c Explain how you would produce loud sounds from this musical instrument.

SKILLS INTERPRETATION (6)

d Draw the trace you might expect to see on an oscilloscope when this instrument is producing a loud, high-pitched note.

SKILLS CRITICAL THINKING (4)

2 a What is an echo?

SKILLS REASONING (5)

b Explain how echoes are used by ships to find the depth of the ocean beneath them.

SKILLS PROBLEM SOLVING (7)

c A ship hears the echo from a sound wave 4 s after it has been emitted. If the speed of sound in water is 1500 m/s, calculate the depth of the water beneath the ship.

SKILLS CRITICAL THINKING (4)

3 a What is meant by the phrase 'a person's audible range is 20 Hz to 20 000 Hz'?

(5)

b Explain why the vibrating strings of a violin produce sounds with a higher frequency than those produced by the strings of a cello.

SKILLS PROBLEM SOLVING (8)

c Calculate the wavelength of sound waves whose frequency is 68 000 Hz. Assume that the waves are travelling through air at a speed of 340 m/s.

4 a An oscilloscope shows a wave which has a time period of 0.01 s. What is the frequency of this wave?

b If the speed of this wave is 340 m/s, calculate its wavelength.

SKILLS CRITICAL THINKING, INTERPRETATION (7)

5 a Sound waves are emitted from a source that is vibrating with a large amplitude and from a source that is vibrating with a small amplitude. Explain, using diagrams, the difference between the two sets of sound waves.

SKILLS INTERPRETATION (6)

b Draw two diagrams to show how these waves would appear on an oscilloscope.

END OF PHYSICS ONLY